SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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COMMON HALF YEARLY EXAMINATION-2024-25

Time Allowed : 3.00 Hours]

MATHEMATICS

[Max. Marks : 90

PART - I

20 x 1 = 20

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. If $|\text{adj}(\text{adj } A)| = |A|^9$, then the order of the square matrix A is

(1) 3 (2) 4 (3) 2 (4) 5

2. The area of the triangle formed by the complex numbers z, iz, and z + iz in the Argand's diagram is

(1) $\frac{1}{2}|z|^2$ (2) $|z|^2$ (3) $\frac{3}{2}|z|^2$ (4) $2|z|^2$

3. According to the rational root theorem, which number is not possible rational zero of $4x^7 + 2x^4 - 10x^3 - 5$?

(1) -1 (2) $\frac{5}{4}$ (3) $\frac{4}{5}$ (4) 5

4. If $\cot^{-1} x = \frac{2\pi}{5}$ for some $x \in R$, the value of $\tan^{-1} x$ is

(1) $-\frac{\pi}{10}$ (2) $\frac{\pi}{5}$ (3) $\frac{\pi}{10}$ (4) $-\frac{\pi}{5}$

5. The radius of the circle passing through the point (6, 2) two of whose diameter are x + y = 6 and x + 2 y = 4 is

(1) 10 (2) $2\sqrt{5}$ (3) 6 (4) 4

6. If the volume of the parallelepiped with $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ as coterminous edges is 8 cubic units, then the volume of the parallelepiped with $(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}), (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})$ and $(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})$ as coterminous edges is,

(1) 8 cubic units (2) 512 cubic units (3) 64 cubic units (4) 24 cubic units

7. The number given by the Mean value theorem for the function $\frac{1}{x}, x \in [1,9]$ is

(1) 2 (2) 2.5 (3) 3 (4) 3.5

8. If $f(x,y,z) = xy + yz + zx$, then $f_x - f_z$ is equal to

(1) z - x (2) y - z (3) x - z (4) y - x

9. If $f(x) = \int_1^x \frac{e^{\sin u}}{u} du, x > 1$ and $\int_1^3 \frac{e^{\sin x^2}}{x} dx = \frac{1}{2}[f(a) - f(1)]$, then one of the possible value of a is

(1) 3 (2) 6 (3) 9 (4) 5

10. The number of arbitrary constants in the general solutions of order n and n +1are respectively

(1) n -1, n (2) n, n +1 (3) n +1, n + 2 (4) n +1, n

11. A computer salesperson knows from his past experience that he sells computers to one in every twenty customers who enter the showroom. What is the probability that he will sell a computer to exactly two of the next three customers?

(1) $\frac{57}{20^3}$ (2) $\frac{57}{20^2}$ (3) $\frac{19^3}{20^3}$ (4) $\frac{57}{20}$

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12. Which one of the following is incorrect? For any two propositions p and q, we have

(1) $\neg(p \vee q) \equiv \neg p \wedge \neg q$ (2) $\neg(p \wedge q) \equiv \neg p \vee \neg q$
(3) $\neg(p \vee q) \equiv \neg p \vee \neg q$ (4) $\neg(\neg p) \equiv p$

13. If the length of the perpendicular from the origin to the plane $2x + 3y + \lambda z = 1, \lambda > 0$ is $\frac{1}{5}$, then the value of λ is

(1) $2\sqrt{3}$ (2) $3\sqrt{2}$ (3) 0 (4) 1

14. The point of inflection of the curve $y = (x-1)^3$ is

(1) (0,0) (2) (0,1) (3) (1,0) (4) (1,1)

15. If $\int_0^a \frac{1}{4+x^2}\,dx = \frac{\pi}{8}$ then a is

(1) 4 (2) 1 (3) 3 (4) 2

16. If focal chord of the parabola $y^2 = ax$ is $2x - y - 8 = 0$ then the equation of directrix is

(1) $x + 4 = 0$ (2) $x - 4 = 0$ (3) $y - 4 = 0$ (4) $y + 4 = 0$

17. The general solution of $\frac{dy}{dx} = \frac{2x-y}{x+2y}$ is –

(1) $x^2 - xy + y^2 = c$ (2) $x^2 - xy - y^2 = c$ (3) $x^2 + xy - y^2 = c$ (4) $x^2 + xy^2 = c$

18. Find $\cos^{-1}\left\{\frac{1}{\sqrt{2}}\left(\cos\frac{9\pi}{10} - \sin\frac{9\pi}{10}\right)\right\} =$

(1) $\frac{3\pi}{20}$ (2) $\frac{7\pi}{20}$ (3) $\frac{7\pi}{10}$ (4) $\frac{17\pi}{20}$

19. $\text{Re}\,\frac{(1+i)^2}{3-i} =$

(1) $-1/5$ (2) $1/5$ (3) $1/10$ (4) $-1/10$

20. Let $f:(0,\infty) \to R$ defined by $f(x) = x + \frac{9\pi^2}{x} + \cos x$ then minimum value of f(x) is

(1) $6\pi - 1$ (2) $10\pi - 1$ (3) $3\pi - 1$ (4) None of these

PART – II

1. **Answer any 7 questions** **$7 \times 2 = 14$**
2. **Each question carries 2 marks**
3. **Question number 30 is compulsory**

21. Find the inverse by Gauss-Jordan method: $\begin{bmatrix} 2 & -1 \\ 5 & -2 \end{bmatrix}$

22. If $z_1 = 3 - 2i$ and $z_2 = 6 + 4i$, find $\frac{z_1}{z_2}$ in the rectangular form

23. Find the value of $\tan(\tan^{-1}(1947))$

24. If the equation of the ellipse is $\frac{(x-11)^2}{484} + \frac{y^2}{64} = 1$ (x and y are measured in centimeters) where to the nearest centimeter should the patient's kidney stone be placed so that the reflected sound hits the kidney stone?

25. Find the asymptotes of the curve : $f(x) = \frac{x^2}{x^2-1}$

26. Find df for $f(x) = x^2 + 3x$ and evaluate it for x = 2 and dx = 0.1

27. Find the domain of the function $\tan^{-1}(\sqrt{9-x^2})$

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28. Find the differential equation of the family of parabolas $y^2 = 4ax$, where a is an arbitrary constant.

29. Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ be any two boolean matrices of the same type. Find $A \vee B$ and $A \wedge B$.

30. If $\vec{a}.\hat{\imath} = \vec{a}.(\hat{\imath} + \hat{\jmath}) = \vec{a}.(\hat{\imath} + \hat{\jmath} + \hat{k})$, then find $\vec{a}$.

PART - III

1. **Answer any 7 questions** **7 x 3 = 21**
2. **Each question carries 3 marks**
3. **Question number 40 is compulsory**

31. If $F(\alpha) = \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix}$, show that $[F(\alpha)]^{-1} = F(-\alpha)$.

32. If $|z| = 2$ show that $3 \leq |z + 3 + 4i| \leq 7$

33. Determine the number of positive and negative roots of the equation $x^9 - 5x^8 - 14x^7 = 0$.

34. Find the equation of the hyperbola passing through $(5, -2)$ and length of the transverse axis along x axis and of length 8 units.

35. Show that the four points $(6, -7, 0), (16, -19, -4), (0, 3, -6), (2, -5, 10)$ lie on a same plane.

36. Find two positive numbers whose sum is 12 and their product is maximum.

37. Find an approximate value of $\int_1^{1.5} (2 - x)dx$ by applying the mid-point rule with the partition $\{1.1, 1.2, 1.3, 1.4, 1.5\}$.

38. Show that $y = 2(x^2 - 1) + Ce^{-x^2}$ is a solution of the differential equation $\frac{dy}{dx} + 2xy - 4x^3 = 0$.

39. Verify (i) closure property (ii) commutative property, and (iii) associative property of the following operation on the given set. $(a * b) = a^b$; $\forall a, b \in N$ (exponentiation property)

40. Write any 3 properties of Cumulative distribution function.

PART - IV

1. **Answer all the questions** **7 x 5 = 35**
2. **Each question carries 5 marks**

41. a) Solve, by Cramer's rule, the system of equations $x_1 - x_2 = 3, 2x_1 + 3x_2 + 4x_3 = 17, x_2 + 2x_3 = 7$

(OR)

b) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$, show that $x^2 + y^2 + 3x - 3y + 2 = 0$.

42. a) Find all zeros of the polynomial $x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$, if it is known that $1 + 2i$ and $\sqrt{3}$ are two of its zeros.

(OR)

b) Solve $\cos\left(\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)\right) = \sin\left\{\cot^{-1}\left(\frac{3}{4}\right)\right\}$

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43. a) Find the eccentricity, center, vertices, foci of $36x^2 + 4y^2 - 72x + 32y - 44 = 0$.

(OR)

b) A ladder 17 metre long is leaning against the wall. The base of the ladder is pulled away from the wall at a rate of 5 m/s. When the base of the ladder is 8 metres from the wall,

(i) how fast is the top of the ladder moving down the wall?

(ii) at what rate, the area of the triangle formed by the ladder, wall, and the floor, is changing?

44. a) Find the local extrema of the function $f(x) = 4x^6 - 6x^4$.

(OR)

b) Let $w(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}, (x,y,z) \neq (0,0,0)$. Show that $\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = 0$.

45. a) Prove by vector method that the perpendiculars (attitudes) from the vertices to the opposite sides of a triangle are concurrent.

(OR)

b) Find the non-parametric form of vector equation and Cartesian equation of the plane passing through the point (1,-2,4) and perpendicular to the plane $x + 2y - 3z = 11$ and parallel to the line $\frac{x+7}{3} = \frac{y+3}{-1} = \frac{z}{1}$.

46. a) Find, by integration, the volume of the solid generated by revolving about y-axis the region bounded between the parabola $x = y^2 + 1$, the y-axis, and the lines $y = 1$ and $y = -1$.

(OR)

b) Suppose a person deposits ₹10,000 in a bank account at the rate of 5% per annum compounded continuously. How much money will be in his bank account 18 months later?

47. a) Two balls are chosen randomly from an urn containing 8 white and 4 black balls. Suppose that we win Rs 20 for each black ball selected and we lose Rs 10 for each white ball selected. Find the expected winning amount and variance.

(OR)

b) (i) Let $M = \left\{\begin{pmatrix} x & x \\ x & x \end{pmatrix} : x \in R - \{0\}\right\}$ and let $*$ be the matrix multiplication. Determine Whether M is closed under $*$. If so, examine the commutative and associative properties satisfied by $*$ on M.

(ii) Let $M = \left\{\begin{pmatrix} x & x \\ x & x \end{pmatrix} : x \in R - \{0\}\right\}$ and let $*$ be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the existence of identity, existence of inverse properties for the operation $*$ on M.

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